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REPLYING TO A SPOKEN COMMAND

Abstract

A method includes, during a teleconference between a first audio input/output device and a second audio input/output device, receiving, at an analysis and response device, a signal indicating a spoken command, the spoken command associated with a command mode. The method further includes, in response to receiving the signal, generating, at the device, a reply message based on the spoken command, the reply message to be output to one or more devices selected based on the command mode. The one or more devices includes the first audio input/output device, the second audio input/output device, or a combination thereof.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This patent application is a continuation of U.S. patent application Ser. No. 18/485,588, filed on Oct. 12, 2023, which is a continuation of U.S. patent application Ser. No. 15/670,572, filed on Aug. 7, 2017, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates generally to replying to spoken commands.

BACKGROUND

[0003] Communication systems may enable users to communicate over large distances. For example, a communication system may be configured to provide a teleconference that enables users in different locations to see and/or hear each other. Such communication systems may be controlled using a remote control device. However, using such a remote control device to control the communication system may be inconvenient. Further, creating a remote control device having a physical button corresponding to each command supported by the communication system may be impractical.

SUMMARY

[0004] Systems and methods according to the disclosure enable a communication system to respond to spoken commands during a communication session between devices. Issuing spoken commands may be more convenient for a user. Further, a manufacturer of a communication system may enable user interactivity with the communication system without providing a remote control that includes a button corresponding to each possible command.

[0005] A method includes, during a teleconference between a first audio input/output device and a second audio input/output device, receiving, at an analysis and response device, a signal indicating a spoken command, the spoken command associated with a command mode. The method further includes, in response to receiving the signal, generating, at the analysis and response device, a reply message based on the spoken command, the reply message to be output to one or more devices selected based on the command mode. The one or more devices includes the first audio input/output device, the second audio input/output device, or a combination thereof.

[0006] An apparatus includes a processor and a memory storing instructions that, when executed by the processor, cause the processor to perform operations. The operations include, during a communication session between a first audio input/output device and a second audio input/output device, receiving a signal indicating a spoken command, the spoken command associated with a command mode. The operations further include, in response to receiving the signal, generating a reply message based on the spoken command, the reply message to be output to one or more devices selected based on the command mode, where the one or more devices include the first audio input/output device, the second audio input/output device, or a combination thereof.

[0007] A computer-readable storage devices stores instructions that, when executed by a processor,

cause the processor to perform operations. The operations include, during a communication session between a first audio input/output device and a second audio input/output device, receiving a signal indicating a spoken command, the spoken command associated with a command mode. The operations further include, in response to receiving the signal, generating a reply message based on the spoken command, the reply message to be output to one or more devices selected based on the command mode, where the one or more devices include the first audio input/output device, the second audio input/output device, or a combination thereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Embodiments described herein are illustrated by way of example and not limitation in the figures of the accompanying drawings, in which like references indicate similar features. It should be understood that the full scope of the inventions disclosed herein are not limited to the precise arrangements, dimensions, and instruments shown. Furthermore, in the drawings, some conventional details have been omitted so as not to obscure the inventive concepts described herein.

[0009] FIG. 1 is a block diagram of a first system for responding to spoken commands operating in a broadcast mode according to the present invention.

[0010] FIG. 2 is another block diagram of the first system for responding to spoken commands operating in a local mode according to the present invention.

[0011] FIG. 3 is block diagram of a second system for responding to spoken commands operating in a broadcast mode according to the present invention.

[0012] FIG. 4 is another block diagram of the second system for responding to spoken commands operating in a local mode according to the present invention.

[0013] FIG. 5 is a flowchart of a method for responding to spoken commands according to the present invention.

[0014] FIG. 6 illustrates a computing device corresponding to a bridge device and operable to perform one or more methods disclosed herein according to the present invention.

[0015] FIG. 7 illustrates a computing device corresponding to communication endpoint device and operable to perform one or more methods disclosed herein according to the present invention.

DETAILED DESCRIPTION

[0016] Reference to the drawings illustrating various views of exemplary embodiments is now made. In the following description, numerous specific details are set forth, such as specific configurations, methods, etc., in order to provide a thorough understanding of the embodiments. At least one of the described embodiments is practicable without one or more of these specific details, or in combination with other known methods and configurations. In other instances, well-known processes and techniques have not been described in particular detail to avoid obscuring the embodiments. Reference throughout this specification to “one embodiment,” “an embodiment,” “another embodiment,” “other embodiments,” “some embodiments,” and their variations means that a particular feature, structure, configuration, or characteristic described in connection with the embodiment is included in at least one implementation. Thus, the appearances of the phrase “in one embodiment,” “in an embodiment,” “in another embodiment,” “in other embodiments,” “in some embodiments,” or their variations in various places throughout this specification are not necessarily referring to the same embodiment. Furthermore, the particular features, structures, configurations, or characteristics are combinable in any suitable manner in one or more embodiments. In the drawings and the description of the drawings herein, certain terminology is used for convenience only and is not to be taken as limiting the embodiments of the present disclosure. Furthermore, in the drawings and the description below, like numerals indicate like elements throughout.

[0017] In the following description and claims, the terms “coupled” and “connected,” along with their derivatives, are not intended as synonyms for each other. “Coupled” is used to indicate that two or more elements or components can directly or indirectly communicate with each other. “Connected” is used to indicate that two or more elements or components are directly linked with each other.

[0018] Any marks that have been referenced herein is by way of example and shall not be construed as descriptive or to limit the scope of the embodiments described herein to material associated only with such marks.

[0019] The present disclosure enables one of skill in the art to provide a system to perform actions in response to spoken commands during a communication session between two devices. In an embodiment according to the present invention, a communication system (e.g., a teleconference system) processes and responds to a spoken command uttered by a participant in a teleconference. In some embodiments, a bridge is configured to facilitate a communication session (e.g., a teleconference) between two devices. In other embodiments, the communication system utilizes peer to peer communications between devices. In embodiments according to the present invention, a device controls, based on a command mode, which devices participating in the communication session output messages associated with the spoken command. For example, in a broadcast mode the command and a reply to the command are inserted into one or more streams of the communication session. In a local mode, the command and the reply are communicated independently of streams of the communication session. FIG. 1 illustrates a system (e.g., a teleconference system) including a bridge device operating in the broadcast mode, and FIG. 2 illustrates the system including the bridge device operating in the local mode. FIG. 3 illustrates a system (e.g., a teleconference system) that utilizes peer to peer connections operating in the broadcast mode, and FIG. 4 illustrates the system that utilizes peer to peer connections operating in the local mode.

[0020] Referring to FIG. 1, a diagram illustrating a system **100** for replying to a spoken command is shown. The system **100** includes a bridge device **102**, a first speaker **108**, a first microphone **110**, a first device (e.g., a first audio input/output device) **104**, a second device (e.g., a second audio input/output device) **106**, a second microphone **112**, and a second speaker **114**. In some implementations, the bridge device **102**, the first device **104**, and the second device **106** correspond to computing devices that each include a corresponding processor and a corresponding memory device (not shown). In some implementations, the first device **104** and the second device **106** correspond to teleconference devices configured to exchange audio data, video data, text data, or a combination thereof. The bridge device **102** is configured to facilitate communication between the first device **104** and the second device **106**. Accordingly, the first device **104** and the second device **106** are endpoints in the communication session.

[0021] The bridge device **102** includes a natural language process backend module **120**, a meeting analytics module **122**, and a natural language process frontend module **124**. The natural language process backend module **120**, the meeting analytics module **122**, and the natural language process frontend module **124** correspond to software executed by a processor of the bridge device **102**, to dedicated hardware of the bridge device **102**, or to a combination thereof. The natural language process frontend module **124** is configured to parse speech or text data to detect a command and to reply to the command. The natural language process backend module **120** is configured to forward the command to another device, retrieve multimedia content, or perform other actions based on the command. The meeting analytics module **122** is configured to generate statistics related to communication (e.g., a teleconference meeting) between the first device **104** and the second device **106**. As used herein, the bridge device **102** is an analysis and response device because the natural language process backend module **120**, the meeting analytics module **122**, and the natural language process frontend module **124** enable the bridge device **102** to analyze and respond to spoken commands.

[0022] The first device **104** is located at a first location **103** and includes a first audio room process module **126** that includes a first wake up detector module **130**, a first echo canceller module **132**, a first mixer module **134**, and a first controller module **136**. The first location **103** usually corresponds to a conference room. The first wake up detector module **130**, the first echo canceller module **132**, the first mixer module **134**, and the first controller module **136** correspond to software executed by a processor of the first device **104**, to dedicated hardware of the first device **104**, or to a combination thereof. Functions of the first wake up detector module **130**, the first echo canceller module **132**, the first mixer module **134**, and the first controller module **136** are described further below. The first device **104** is coupled to the first speaker **108** and the first microphone **110**. In some embodiments, the first device **104** is further coupled to a first display device (not shown). In some implementations of the system **100**, one or both of the first speaker **108**, the first microphone **110**, and the first display device are components of the first device **104** rather than separate devices. In some implementations of the system **100**, the first device **104** include more or fewer components than illustrated.

[0023] The second device **106** is located at a second location **105** and includes a second audio room process module **128** that includes a second echo canceller module **138**, a second wake up detector module **140**, a second controller module **142**, and a second mixer module **144**. The second location **105** is distinct from the first location **103** and in some embodiments corresponds to a conference room. The second wake up detector module **140**, the second echo canceller module **138**, the second mixer module **144**, and the second controller module **142** correspond to software executed by a processor of the second device **106**, to dedicated hardware of the second device **106**, or to a combination thereof. Functions of the second wake up detector module **140**, the second echo canceller module **138**, the second mixer module **144**, and the second controller module **142** are described further below. The second device **106** is coupled to the second speaker **114** and the second microphone **112**. In some embodiments, the second device **106** is further coupled to a second display device (not shown). In some implementations of the system **100**, one or both of the second speaker **114**, the second microphone **112**, and the second display device are components of the second device **106** rather than separate devices. In some implementations of the system **100**, the second device **106** includes more or fewer components than illustrated.

[0024] The bridge device **102** supports a communication session (e.g., a teleconference) between the first device **104** and the second device **106**. The communication session is conducted via a plurality of streams. In the example shown in FIG. 1, the communication session is conducted via a first input stream **146**, a first output stream **148**, a second input stream **150**, and a second output stream **152**. Each of the plurality of streams carries data corresponding to multimedia content. For example, the first output stream **148** carries data corresponding to audio or video content captured at the first location **103** (e.g., by first microphone **110** and/or a video capture device) from the first device **104** to the bridge device **102**. The bridge device **102** generates the second input stream **150** based on the first output stream **148** and transmit the second input stream **150** to the second device **106**. Similarly, the second output stream **152** carries data corresponding to audio or video content captured at the second location **105** (e.g., by the second microphone **112** and/or a video capture device) from the second device **106** to the bridge device **102**. The bridge device **102** generates the first input stream **146** based on the second output stream **152** and transmits the first input stream **146** to the first device **104**. Thus, the bridge device enables the communication session between the first device **104** and the second device **106**. The communication session enables a first user **160** at the first location **130** to speak with and/or see a second user **162** at the second location **105**.

[0025] During the communication session, the bridge device **102** receives and responds to data indicating one or more spoken commands from one or both of the first device **104** and the second device **106**. The bridge **102** is configured to respond to a particular command based on a command mode associated with the particular command. FIG. 1 illustrates an example of the bridge device **102** responding to a command in a broadcast mode. In some implementations, the bridge device

102 also respond to commands according to a local mode, as described below with reference to FIG. 2. Further, in some implementations, the bridge device **102** is configured to operate in the local mode with respect to commands from one device while operating in broadcast mode with respect to commands from the other device.

[0026] In the example illustrated in FIG. 1, the first user **160** speaks a wake up phrase **166**, and the first microphone **110** receives acoustic waves corresponding to a wake up phrase **166'** spoken by the first user **160** during the communication session (e.g., during a teleconference). For example, the first microphone **110** receives acoustic waves corresponding to the first user **160** saying "wake up." The first microphone **110** converts the acoustic waves corresponding to the wake up phrase **166'** to a signal (e.g., a digital or analog electrical or radio signal) indicating the wake up phrase **166''** and transmits the signal indicating the wake up phrase **166''** to the first device **104**. The first wake up detector module **130** searches input from the first microphone **110** for the wake up phrase **166**. In response to receiving the signal indicating the wake up phrase **166''**, the first wake up detector module **130** causes the first device to transmit a wake up message **170** to the bridge device **102**. The wake up message **170** indicates a command mode **172**. The first wake up detector module **130** determines the command mode **172** based on the wake up phrase **166**. In an illustrative example, the first wake up detector module **130** sets the command mode **172** to broadcast mode in response to the wake up phrase **166** corresponding to "wake up broadcast." It should be noted that in other embodiments, the first device **104** transmits the wake up message **170** in response to a different triggering signal. In an illustrative example, the first device **104** transmits the wake up message **170** to the bridge device **102** in response to a signal from a remote control device indicating a button selection. In some embodiments, the first device **104** selects the command mode **172** based on content of the triggering signal, a source (e.g., the remote control device or the first microphone **110**) of the triggering signal, or based on a default setting. In an illustrative example, the first device **104** selects the command mode **172** based on which button of the remote control device is indicated by the signal (or based on content of the wake up phrase **166**). In another illustrative example, the first device **104** sets the command mode **172** to the broadcast mode based on a default setting in response to the triggering signal not indicating a command mode selection. [0027] In response to receiving the wake up message **170**, the natural language process frontend module **124** "wakes up" and begin searching one or more streams for one or more spoken commands based on the command mode **172**. In the broadcast mode, the natural language process frontend module **124** searches streams associated with the communication session (e.g., the teleconference) for the one or more commands.

[0028] In addition to outputting the wake up message **170**, the first device **104** includes data corresponding to the wake up phrase **166''** in the first output stream **148**. For example, the first controller module **136** inserts the data corresponding to the wake up phrase **166'''** into the first output stream **148**. The bridge device **102** receives the data corresponding to the wake up phrase **166''** and, in response to the command mode **172** indicating the broadcast mode, includes the data corresponding to the wake up phrase **166'''** in the second input stream **150**. The second device **106** receives the data corresponding to the wake up phrase **166''**, and the second mixer module **144** outputs an audio signal corresponding to the wake up phrase **166crv**) to the second speaker **114**. In response to the audio signal corresponding to the wake up phrase **166crv**), the second speaker **114** outputs acoustic waves corresponding to the wake up phrase **166CV**) at the second location **105**. Further, the second microphone **112** detects the acoustic waves corresponding to the wake up phrase **166(V)** and transmits an audio signal indicating the wake up phrase **166cvr**) to the second device **106**.

[0029] In addition to transmitting the audio signal indicating the wake up phrase **166cvr**) to the second speaker **114**, the second mixer module **144** transmits the audio signal indicating the wake up phrase **166cvr**) to the second echo canceller module **138**. Based on the audio signal indicating the wake up phrase **166(VI)**, the second echo canceller module **138** removes the wake up phrase

166 from the audio signal from the second microphone **112**. Accordingly, the second wake up detector module **140** does not detect the wake up phrase **166** and data corresponding to the wake up phrase **166** is not included in the second output stream **152**.

[0030] As described above, the natural language process frontend module **124** of the bridge device **102** searches streams of the communication session for one or more spoken commands in response to the command mode **172** indicating the broadcast mode. The one or more spoken commands include a request to create a summary of the communication session, a request for information (e.g., weather forecast), a request for multimedia content, a request to control a device, another command, or a combination thereof.

[0031] In the example illustrated in FIG. **1**, the first user **160** speaks a command **164** during the communication session, and the first microphone **110** receives acoustic waves corresponding to the command **164'**. In an illustrative example, the first microphone **110** receives acoustic waves corresponding to the first user **160** saying "What's the weather like?" The first microphone **110** converts the acoustic waves corresponding to the command **164'** to a signal (e.g., a digital or analog electrical or radio signal) indicating the command **164''** and transmits the signal indicating the command **164''** to the first device **104**. Based on the signal indicating the command **164''**, the first device **104** inserts data corresponding to the command **164'''** into the first output stream **148**. In a particular embodiment, the first controller module **136** inserts the data corresponding to the command **164'''** into the first output stream **148**.

[0032] As used herein a spoken command, such as the command **164**, includes a request for information, a request to control a part of the communication session, a request for a media content item, a request to control a device, another type of command, or a combination thereof. A request to control a part of the communication session includes a request to mute/unmute one or more devices at a particular site, a request to mute/unmute a device associated with the requester, a request to mute/unmute all devices except for a device associated with the requester, a request to connect to another device, a request to share/stop sharing multimedia content during the communication session, a request to show/hide video associated with the requester, a request to change a presentation mode associated with the communication session, another type of command, or a combination thereof. A request to control a device includes a request to activate the device, a request to deactivate the device, a request to adjust a setting (e.g., volume, brightness, etc.) of the device, another type of request, or a combination thereof.

[0033] The natural language process frontend module **124** detects the data corresponding to the command **164'''** in the first output stream **148**, as described further below. Additionally, in response to the command mode **172** indicating the broadcast mode, the bridge device **102** includes the data corresponding to the command **164'''** in the second input stream **150**. The second device **106** receives the data corresponding to the command **164'''**. Based on the data corresponding to the command **164''**, the second mixer module **144** outputs an audio signal corresponding to the command **164crv**) to the second speaker **114**, and the second speaker **114** outputs acoustic waves corresponding to the command **164CV**) at the second location **105** by the second speaker **114**. Further, the second microphone **112** detects the acoustic waves corresponding to the command **164CV**) and transmits an audio signal indicating the command **164cvr**) to the second device **106**. In addition, the second echo canceller module **138** may perform echo cancellation, as described above, to remove the command from the audio signal from the second microphone **112**. Accordingly, data corresponding to the command **164** is not be included (or is lessened) in the second output stream **152**.

[0034] As explained above, the natural language process frontend module **124** detects the data corresponding to the command **164''** in the first output stream **148**. The natural language process frontend module **124** performs one or more actions based on content of the command **164**, based on the command mode **172**, based on a communication session context, or a combination thereof. In some embodiments, the communication session context is identified by the meeting analytics

module **122** and includes information identifying one or more “meeting objects.” The one or more meeting objects include a speaker of the command **164**, a list of participants in the communication session, a list of locations associated with the communication session, a list of user profiles associated with the communication session, a list of available commands, a list of media content items associated with the communication session, a list of devices associated with the communication session, a list of supported spoken commands, or a combination thereof.

[0035] In some examples, the meeting analytics module **122** generates a graphical user interface based on the communication session context, one or more contexts associated with previous communication sessions, or a combination thereof. The graphical user interface identifies one or more meeting objects. The bridge device **102** transmits the graphical user interface to the first device **104** and the second device **106** as part of the communication session (e.g., via the first input stream **146** and the second input stream **150**). The first device **104** and the second device **106** output (e.g., to display devices) the graphical user interface to enable users to interact with the graphical user interface using one or more spoken commands (e.g., the command **164**). In a particular embodiment, the meeting analytics module **122** monitors the first output stream **148** and the second output stream **152** to identify a keyword spoken during the communication session. In some embodiments, a keyword corresponds to a tag applied (e.g., by a user or by an automated system) to one or more media content items accessible to the natural language process backend module **120**. In response to detecting the keyword in the communication session, the first meeting analytics module **122** causes the graphical user interface to display the keyword. In some embodiments, the command **164** corresponds to a request for the one or more media content items associated with the keyword.

[0036] In a first example of performing one or more actions, in response to determining that content of the command **164** corresponds to the phrase “create a meeting summary,” the natural language process frontend module **124** causes the meetings analytics module **122** to create a meeting summary associated with the communication session. Creating the meeting summary includes recording the communication session, generating a speech to text transcription based on the recording, identifying key phrases from the transcription, identifying sentiments expressed in the communication session, identifying active speakers in the communication session, identifying communication session participants, generating minutes of the communication session, or a combination thereof. In some embodiments, the meeting analytics module **122** further causes the natural language process backend module **120** to E-mail the summary to meeting participants based on the context session context.

[0037] In a second example, in response to determining that content of the command **164** corresponds to the phrase “what’s the weather like?” the natural language process frontend module **124** identifies that the command **164** is associated with the first location **103** based on the communication session context. The natural language process frontend module **124** causes the natural language process backend module **120** to transmit a request for weather information to a weather service.

[0038] In a third example, in response to determining that content of the command **164** corresponds to the phrase “turn the lights on” the natural language process frontend module **124** identifies that the command **164** is associated with the first location **103** based on the communication session context (e.g., the command being from a user at the first location **103** and being a command that specifies a location-centric operation, such as controlling a lighting device). The natural language process frontend module **124** then causes the natural language process backend module **120** to transmit an activation signal to a lighting device associated with the first location **103**. In some embodiments, the first user **160** provides an additional phrase to override context-based location processing. For example, in response to the command **164** corresponding to the phrases “turn the lights on” and “at the second location,” the natural language process frontend module **124** identifies that the command **164** is associated with the second location **106**.

[0039] In a fourth example, the natural language process frontend module **124** determines whether the first user **160** is authorized to initiate the command **164** (e.g., muting another user). The natural language process frontend module **124** determines whether the first user **160** is authorized based on a user profile accessible to the meeting analytics module **122**. In an illustrative example, the user profile indicates a permission level associated with the first user **160**, and the natural language process frontend module **124** determines whether the first user **160** is authorized to initiate the command **164** based on the permission level.

[0040] In some implementations, each permission level is associated with set of privileges that define what commands may be initiated by a user having the permission level. Examples of permission levels include “Administrator,” “Presenter,” and “Listener.” Examples of privileges include “granted,” “prohibited,” and “by request.” Users associated with a permission level are able to initiate all “granted” commands associated with the permission level and are not able to initiate “prohibited” commands associated with the permission level. Users of a permission level must receive permission from another user (e.g., a user associated with the “Administrator” permission level) before initiating “by request” commands associated with the permission level.

[0041] In an illustrative example, the command **164** is a by request command associated with a profile of the first user **160** and the second user **162** is an administrator. Accordingly, the natural language process frontend module **124** transmits a permission request to the second user **162** in response to the command **164**. The second device **106** generates a response to the permission request based on input (e.g., voice input) from the second user **162**. The second device **106** receives the input from the second user **162** either before or after the permission request is sent to the second device **106**. To illustrate, the second device **106** may output the permission request via the second speaker **114** and then the second user **162** may enter the input via the second microphone **112** (e.g., by speaking “yes” or “no”). Alternatively, the second user **162** may enter the input via the second microphone **112** (e.g., by speaking “automatically grant permission to listeners to initiate X” or “automatically deny permission to listeners to initiate X” where X is a command) without the second device **106** outputting the permission request. Based on the response to the permission request, the natural language process frontend **124** determines whether the first user **160** is authorized to initiate the command **164**.

[0042] In a fifth example, a presenter discussing a trip to Rome during the communication session says a keyword, such as “Rome.” The meeting analytics module **122** determine that the natural language process backend module **120** has access (e.g., via a storage device of the bridge device **102** or via a network connection) to an image tagged with the keyword (e.g., “Rome”).

Accordingly, the meeting analytics module **122** updates the graphical user interface to display the keyword, and the first device **104** displays the user interface to the first user **160**. In this example, the command **164** corresponds to the first user **160** saying “Show me more about the Rome trip,” and the one or more actions include retrieving the media content item associated with the keyword “Rome.”

[0043] After performing the one or more actions, the natural language process frontend module **124** generates a reply **168**. In the illustrated example, the reply **168** corresponds to audio content (e.g., generated using a text to speech process), but in some embodiments, the reply **168** includes audio content, video content, text content, or any combination thereof. The reply **168** indicates information (e.g., a weather report) retrieved as part of the one or more actions, reports that the one or more actions have been completed (e.g., “the lights are now on”), reports that the first user **160** is not authorized to initiate the command **164** (e.g., based on the user profile), includes a media content item (e.g., associated with a keyword, as described above), or a combination thereof. In response to the command mode **172** indicating the broadcast mode, the natural language process frontend module **124** broadcasts data corresponding to the reply **168** as part of the communication session.

[0044] The first device **104** receives the data corresponding to the reply **168**, and the first mixer

module **134** outputs an audio signal corresponding to the reply **168''** to the first speaker **108**. Based on the audio signal corresponding to the reply **168''**, the first speaker **108** outputs acoustic waves corresponding to the reply **168''** at the first location **103**. Further, the first microphone **110** detects the acoustic waves corresponding to the reply **168'''** and transmits an audio signal indicating the reply **168crv**) to the first device **104**. The first echo canceller **132** removes or reduces the reply **168** from the audio signal, as explained above. Accordingly, data corresponding to the reply **168** is not included (or is reduced) in the first output stream **148**.

[0045] Further, the second device **106** receives the data corresponding to the reply **168'**. The second mixer module **144** outputs an audio signal corresponding to the reply **168(V)** to the second speaker **114**. Based on the audio signal corresponding to the reply **168CV** the second speaker **114** outputs acoustic waves corresponding to the reply **168cvr**) at the second location **105**. Further, the second microphone **112** detects the acoustic waves corresponding to the reply **168cvr**) and transmits an audio signal indicating the reply **168cvn**) to the second device **106**. The second echo canceller **138** removes or reduces the reply **168** from the audio signal received from the second microphone **112**, as explained above. Accordingly, data corresponding to the reply **168** is not included (or is reduced) in the second output stream **152**.

[0046] It should be noted that in some use cases, the bridge device **102** receives a wake up message from one device and respond to a spoken command received from a different device. For example, in some implementations, after receiving the wake up message **170** from the first device **104**, the bridge device **102** receives a spoken command received from the second device **106**. In some embodiments, the bridge device **102** is configured to respond to any detected spoken command regardless of a source of the spoken command. Accordingly, in some embodiments, the bridge device **102** responds to the spoken command received from the second device **106** after receiving the wake up message **170** from the first device **104**. In some embodiments, the wake up message **170** indicates to which devices the bridge **102** should or should not respond. Thus, in some embodiments, the bridge device **102** is configured to, based on the wake up message **170**, respond to spoken commands received from one or more devices and to ignore spoken commands received from one or more other devices. Accordingly, in some embodiments, the bridge device **102** ignores the spoken command received from the second device **106** based on the wake up message **170** indicating that the bridge device **102** should process spoken commands from the first device **104** or that the bridge device **102** should not process spoken commands from the second device **106**. In some examples, the first device **104** causes the wake up message **170** identifies devices based on the wake up phrase **166**, based on button commands indicated by a signal from a remote control device, or a combination thereof.

[0047] In some implementations, the natural language process frontend module **124** “goes back to sleep” (e.g., ceases monitoring the first output stream **148** and the second output stream **152**) in response to generating a reply (e.g., the reply **168**), in response to a threshold period of time elapsing since receipt of the wake up message **170**, or in response to a sleep message received from the first device **104** (or the second device **106**). In some embodiments, the sleep message is generated by the first audio room process module **126** (or the second audio room process module **128**) in response to a sleep trigger (e.g., passing of a threshold period of time since generation of the wake up message **170**, detection of a sleep phrase in signals generated by the first microphone **110** (or the second microphone **112**), receipt of a signal from a remote control device, etc.).

[0048] Thus, the system **100** of FIG. **1** enables processing of and response to voice commands during a communication session. In particular, the system **100** determines to transmit a command spoken during a communication session to one or more devices selected based on a command mode associated with the command. Further, the system **100** transmits a reply to the command to devices selected based on the command mode. As illustrated in FIG. **1**, the first device **104** and the bridge device **102**, operating according to a broadcast mode, place command content and reply content in streams associated with a communication session. Accordingly, a command spoken at

the first location **103** and during the communication session is presented at a second location participating in the communication session. Further, a reply to the command is presented at the second location.

[0049] Referring to FIG. 2, another diagram illustrating the system **100** for replying to a spoken command is shown. As indicated above, FIG. 2 illustrates the bridge device **102** responding to a spoken command according to a local mode. As explained above, the bridge device **102** enables the communication session between the first device **104** and the second device **106** via the first input stream **146**, the first output stream **148**, the second input stream **150**, and the second output stream **152**.

[0050] In the example illustrated in FIG. 2, the first user **160** speaks a wake up phrase **202**, and the first microphone **110** receives audio waves corresponding to the wake up phrase **202'** during the communication session (e.g., during a teleconference). For example, the first microphone **110** receives audio waves corresponding to the first user **160** saying "wake up local mode." The first microphone **110** converts the audio waves corresponding to the wake up phrase **202'** to a signal (e.g., a digital or analog electrical or radio signal) indicating the wake up phrase **202"** and transmits the signal to the first device **104**. The first wake up detector module **130** searches input from the first microphone **110** for the wake up phrase **202**. In response to detecting the signal indicating the wake up phrase **202"**, the first wake up detector module **130** causes the first device **104** to transmit a wake up message **204** to the bridge device **102**. The wake up message **204** indicates a command mode **206**. The first wake up detector module **130** determines the command mode **206** based on the wake up phrase **202**. For example, in some implementations, the first wake up detector module **130** sets the command mode **206** to local mode in response to the wake up phrase **202** corresponding to "wake up local mode." It should be noted that in other embodiments, the first device **104** transmits the wake up message **204** in response to a different triggering signal. For example, in some embodiments, the first device **104** transmits the wake up message **204** to the bridge device **102** in response to a signal from a remote control device indicating a button selection. Further, in some embodiments, the first device **104** selects the command mode **206** based on content of the triggering signal or a source (e.g., the remote control device or the first microphone **110**) of the triggering signal. For example, the first device **104** selects the command mode **206** based on which button of the remote control device is indicated by the signal (or based on content of the wake up phrase **202**). In some embodiments, the wake up phrase **202**, the button (or a combination of buttons) indicated by the signal, or a combination thereof further identify to which of the first device **104** and the second device **106** the bridge device **102** is to respond in the local mode.

[0051] In addition to transmitting the wake up message **103** to the bridge device **102**, the first device **104** establishes additional communication streams with the bridge device **102** in response to the triggering signal (e.g., the wake up phrase **202** or the remote control signal). In some examples, the first device **104** establishes the additional communication streams in response to the wake up phrase **202**, the button (or combination of buttons) indicated by the signal from the remote control device, or a combination thereof identifying the first device **104**. In alternative embodiments (not shown), the wake up message **204** instructs the bridge device **102** to establish the additional communication streams with the second device **106** in response to the wake up phrase **202**, the button (or combination of buttons) indicated by the signal from the remote control device, or a combination thereof identifying the second device **106**.

[0052] In the example illustrated in FIG. 2, the first audio room process module **126** establishes a local input stream **210** to receive data from the bridge device and a local output stream **208** to transmit data to the bridge device. In some examples, the wake up message **204** is transmitted to the bridge device **102** via the local output stream **208**. The local input stream **210** and the local output stream **208** are distinct from the communication session. For example, the bridge device **102** does not transmit content received via the local output stream **208** to the second device **106**. The first controller module **136** controls which of the first output stream **148** and the local output stream

208 is used to output content to the bridge device **102**. In some examples, the first controller module **136** “mutes” (e.g., ceases transmitting content through) the first output stream **148** in response to the command mode **206** corresponding to the local mode. Accordingly, the second device **106** does not receive content from the first device **104** while the first device **104** and the bridge device **102** are operating in the local mode. The first mixer module **134** generates audio signals to be output by the first speaker **108** based on one or both of the local input stream **210** and the first input stream **146**. In some examples, the first mixer module **134** “mutes” (e.g., cease using) the first input stream **210** while the first device **104** and the bridge device **102** are operating in the local mode.

[0053] In response to receiving the wake up message **204**, the natural language process frontend module **124** “wakes up” and begins searching one or more streams for one or more spoken commands based on the command mode **206**. In the local mode, the natural language process frontend module **124** searches one or more streams that are not associated with the communication session (e.g., the teleconference) for the one or more commands. In the example illustrated in FIG. 2, the natural language process frontend module **124** searches the local output stream **208** for the one or more commands in response to the command mode **206** corresponding to the local mode. As explained above, the one or more spoken commands includes a request to create a summary of the communication session, a request for information (e.g., weather forecast), a request for multimedia content, a request to control a device, another type of command, or a combination thereof.

[0054] In the example illustrated in FIG. 2, the first user **160** speaks a command **212**, and the first microphone **110** receives audio waves corresponding to the command **212'** spoken by the first user **160** during the communication session (e.g., during a teleconference). In a particular example, the first microphone **110** receives audio waves corresponding to the first user **160** saying “What's the weather like?” The first microphone **110** converts the audio waves corresponding to the spoken command **212'** to a signal (e.g., a digital or analog electrical or radio signal) indicating the command **212''** and transmits the signal indicating the command **212''** to the first device **104**. Based on the signal indicating the command **212''**, the first device **104** generates data corresponding to the command **212''**. In response to the command mode **206** corresponding to the local mode, the first controller module **136** inserts the data corresponding to the command **212'''** into the local output stream **208** but not into the first output stream **148**.

[0055] The natural language process frontend module **124** detects the data corresponding to the command **212'''** in the local output stream **208**, and performs one or more actions. The one or more actions are based on content of the command **212**, the command mode **206**, a communication session context, or a combination thereof. In some embodiments, the communication session context is identified by the meeting analytics module **122** and includes information identifying one or more “meeting objects.” The one or more meeting objects include a speaker of the command **212**, a list of participants in the communication session, a list of locations associated with the communication session, a list of user profiles associated with the communication session, a list of available commands, a list of media content items associated with the communication session, a list of devices associated with the communication session, a list of supported spoken commands, or a combination thereof.

[0056] In some examples, the meeting analytics module **122** generates a graphical user interface based on the communication session context, one or more contexts associated with previous communication sessions, or a combination thereof. The graphical user interface identifies one or more meeting objects. The bridge device **102** transmits the graphical user interface to the first device **104** and the second device **106** as part of the communication session (e.g., via the first input stream **146** and the second input stream **150**). The first device **104** and the second device **106** output (e.g., to display devices) the graphical user interface to enable users to interact with the graphical user interface using one or more spoken commands (e.g., the command **212**). In a particular embodiment, the meeting analytics module **122** monitors the first output stream **148** and

the second output stream **152** to identify a keyword spoken during the communication session. In some embodiments, a keyword corresponds to a tag applied (e.g., by a user or by an automated system) to one or more media content items accessible to the natural language process backend module **120**. In response to detecting the keyword in the communication session, the first meeting analytics module **122** causes the graphical user interface to display the keyword. In some embodiments, the command **212** corresponds to a request for the one or more media content items associated with the keyword.

[0057] After performing the one or more actions, the natural language process frontend module **124** generates a reply **214**. In the illustrated example, the reply **214** corresponds to audio content (e.g., generated using a text to speech process), but in some embodiments, the reply **214** includes audio content, video content, text content, or any combination thereof. The reply **214** indicates information (e.g., a weather report) retrieved as part of the one or more actions, reports that the one or more actions have been completed (e.g., “the lights are now on”), reports that the first user **160** is not authorized to initiate the command **212** (e.g., based on the user profile), includes a media content item (e.g., associated with a keyword, as described above), or a combination thereof. In response to the command mode **206** indicating the local mode, the natural language process frontend module **124** transmits data corresponding to the reply **214** via the local input stream **210** but not through streams associated with the communication session.

[0058] The first device **104** receives the data corresponding to the reply **214**. In response to the data corresponding to the reply **214**, the first mixer module **134** outputs an audio signal corresponding to the reply **214** to the first speaker **108**, and based on the audio signal corresponding to the reply **214**, the first speaker **108** outputs acoustic waves corresponding to the reply **214**. Further, the first microphone **110** detects the acoustic waves corresponding to the reply **214** and transmits an audio signal indicating the reply **214** to the first device **104**. As explained above, the first echo canceller module **132** removes or reduces the reply **214** in the audio signal received from the first microphone **110**. Accordingly, data corresponding to the reply **214** is not included (or is reduced) in the local output stream **208**.

[0059] In some implementations, the natural language process frontend module **124** “goes back to sleep” (e.g., cease monitoring the local output stream **208**) in response to generating a reply (e.g., the reply **214**), in response to a threshold period of time elapsing since receipt of the wake up message **204**, or in response to a sleep message received from the first device **104**. In some embodiments, the sleep message is generated by the first audio room process module **126** in response to a sleep trigger (e.g., passing of a threshold period of time since generation of the wake up message **204**, detection of a sleep phrase in signals generated by the first microphone **110**, receipt of a signal from a remote control device, etc.). The first controller module **136** “unmutes” the first output stream **148** and mutes the local output stream **208** in response to determining (e.g., based on a notification from the first audio room process module **126** or the natural language process frontend module **124**) that the natural language process frontend module **124** is asleep. Accordingly, the second device **106** receives content from the first device **104** via the bridge device **102** after the natural language process front end module **125** “goes to sleep.”

[0060] Thus, the system **100** of FIG. 2 enables processing of and response to voice commands during a communication session. In particular, the system **100** determines to transmit a command spoken during a communication session to one or more devices selected based on a command mode associated with the command. Further, the system **100** transmits a reply to the command to devices selected based on the command mode. As illustrated in FIG. 2, the first device **104** and the bridge device **102** operating according to a local mode establish communication streams separate from the communication session. Accordingly, a command spoken at the first location **103** and during a communication session is not presented at a second location participating in the communication session. Further, a reply to the command is not presented at the second location.

[0061] It should be noted that, in some embodiments, the bridge device **102** operates according to

the broadcast mode with respect to one device while operating according to the local mode with respect to another device. In an illustrative example, the bridge device **102** responds to a first spoken command from the first device **104** according to the local mode (as illustrated in FIG. 2) and a second spoken command from the second device **106** according to the broadcast mode (as illustrated in FIG. 1). Thus, the second command is transmitted to the first device **104** as part of a communication session between the first device **104** and the second device **106**, and a reply to the second spoken command is broadcast to both the first device **104** and the second device **106** as part of the communication session. Further, a reply to the first command is transmitted to the first device **104** via a channel distinct from the communication session.

[0062] It should further be noted that, in some embodiments, the bridge device **102** supports additional operations modes. In an illustrative private mode, the bridge device **102** receives a text message from a third device via a data stream distinct from the communication session between the first device **104** and the second device **106**. The bridge device **102** performs one or more actions, as explained above, in response to one or more commands included in the text message. The bridge device **102** further transmits a reply message (e.g., a reply text message) to the third device. In some implementations, the third device participates in the communication session between the first device **104** and the second device **106**. In other implementations, the third device does not participate in the communication session.

[0063] Referring to FIG. 3, a diagram illustrating a system **300** for replying to a spoken command during a peer to peer communication session is shown. The system **300** includes a first device **304** located at a first location **303** and a second device **306** located at a second location **305**. The first device **304** and the second device **306** correspond to the first device **104** and the second device **106** of the system **100** respectively. Accordingly, the first device **304** includes a first audio room process module **326** that corresponds to the first audio room process module **126** and the second device **306** includes a second audio room process module **328** that corresponds to the second audio room process module **128**. The first audio room process module **326** includes a first wake up detector module **330**, a first echo canceller module **332**, a first mixer module **334**, and a first controller module **336** that correspond to the first wake up detector module **130**, the first echo canceller module **132**, the first mixer module **134**, and the first controller module **136** respectively. The second audio room process module **328** includes a second wake up detector module **340**, a second echo canceller module **338**, a second mixer module **344**, and a second controller module **342** that correspond to the second wake up detector module **140**, the second echo canceller module **138**, the second mixer module **144**, and the second controller module **142** respectively. In addition, the first device **304** and the second device **306** include components and/or functionality of the bridge device **102**. In the illustrated example, the first device **304** includes a first natural language process backend module **320**, a first meeting analytics module **322**, and a first natural language process frontend module **324** that correspond to the natural language process backend module **120**, the meeting analytics module **122**, and the natural language process frontend module **124**. The first natural language process backend module **320**, the first meeting analytics module **322**, and the first natural language process frontend module **324** correspond to software executable by a processor of the first device **304**, to dedicated hardware components, or to a combination thereof. The second device **306** includes a second natural language process backend module **370**, a second meeting analytics module **372**, and a second natural language process frontend module **374** that correspond to the natural language process backend module **120**, the meeting analytics module **122**, and the natural language process frontend module **124**. The second natural language process backend module **370**, the second meeting analytics module **372**, and the second natural language process frontend module **374** correspond to software executable by a processor of the second device **306**, to dedicated hardware components, or to a combination thereof. Accordingly, the first device **304** and the second device **306** are analysis and response devices. In some implementations, only one of the first device **304** and the second device **306** is an analysis and response device. The first device **304**

is coupled to a first speaker **308** and a first microphone **310** that correspond to the first speaker **108** and the first microphone **110**. The second device **306** is coupled to a second speaker **314** and a second microphone **312** that correspond to the second speaker **114** and the second microphone **112**. [0064] The first device **304** and the second device **306** communicate in a peer to peer communication session conducted over a first stream **346** and a second stream **348**. FIG. 3 illustrates the system **300** operating according to the broadcast mode, as described above with reference to FIG. 1. Accordingly, a wake up phrase **366** and a command **364** spoken by a first user **360** at the first location **303** are received at the first device **304** and transmitted to the second device **306** as part of the communication session. Further, a reply **368** generated by the first natural language process frontend module **324** in response to the command **364**, as described above with reference to FIG. 1, is transmitted to the first device **304** (e.g., from one component of the first device **304** to another component of the first device **304**) and to the second device **306** as part of the peer to peer communication session. Accordingly, both the first speaker **308** and the second speaker **314** output the reply **368** and the second speaker **314** outputs the wake up phrase **366** and the command **364**.

[0065] Referring to FIG. 4, another diagram illustrating the system **300** for replying to a spoken command is shown. As indicated above, FIG. 4 illustrates the first device **304** responding to a spoken command according to the local mode, as described with reference to FIG. 2. Accordingly, a wake up phrase **466** and a command **464** spoken by the first user **360** at the first location **303** are received at the first device **304** and are not transmitted to the second device **306** as part of the communication session. Further, a reply **468** generated by the first natural language process frontend module **324** in response to the command **464**, as described above with reference to FIG. 2, is transmitted to the first device **304** (e.g., from one component of the first device **304** to another component of the first device **304**) but is not transmitted to the second device **306** as part of the peer to peer communication session. Accordingly, the first speaker **308** outputs the reply **468**, and the second speaker **314** does not output the wake up phrase **466**, the command **464**, or the reply **468**.

[0066] It should be noted that in some embodiments the first device **304** operates according to the broadcast mode while the second device **306** operates according to the local mode. In an illustrative example, the first device **304** responds to a first spoken command according to the local mode (as illustrated in FIG. 4), and the second device **306** responds to a second spoken command according to the broadcast mode (as illustrated in FIG. 3). Thus, the second command is transmitted to the first device **104** as part of a communication session between the first device **304** and the second device **306**, and a reply to the second spoken command is broadcast to both the first device **304** and the second device **306** as part of the communication session. Further, a reply to the first command is not included in the communication session.

[0067] Referring to FIG. 5, a flowchart illustrating a method **500** of replying to a spoken command is shown. In some embodiments, the method **500** is performed by the bridge device **102** of FIGS. 1 and 2, the first device **304** of FIGS. 3 and 4, the second device **306** of FIGS. 3 and 4, or a combination thereof.

[0068] The method **500** includes, during a communication session between a first device and a second device, receiving a wake up message, at **502**. In the example of FIG. 1, the bridge device **102** receives the wake up message **170**. In the example of FIG. 2, the bridge device receives the wake up message **204**. In the example, of FIG. 3, the first device **304** receives a signal indicating the wake up phrase **366**". In the example of FIG. 4, the first device **304** receives a signal indicating the wake up phrase **466**".

[0069] The method **500** further includes determining whether a command mode a broadcast mode or a local mode, at **504**. In the example of FIG. 1, the bridge device **102** determines that the command mode **172** associated with the wake up message **170** is the broadcast mode. In the example of FIG. 2, the bridge device **102** determines that the command mode **206** associated with

the wake up message **204** is the local mode. In the example of FIG. **3**, the first device **304** determines that a command mode associated with the wake up phrase **366** is the broadcast mode (e.g., based on content of the wake up phrase **366**, based on a signal from a remote control device, based on a context of the communication session, or combination thereof). In the example of FIG. **4**, the first device **304** determines that a command mode associated with the wake up phrase **466** is the local mode (e.g., based on content of the wake up phrase **366**, based on a signal from a remote control device, based on a context of the communication session, or combination thereof).

[0070] In response to determining that command mode is the broadcast mode, the method **500** further includes broadcasting a wake up phrase as part of the communication session, at **506**. In the example of FIG. **1**, the bridge device **102** broadcasts the data corresponding to the wake up phrase **166'''** via the second input stream **150** as part of the communication session between the first device **104** and the second device **106**. In the example of FIG. **3**, the first device **304** broadcasts the data corresponding to the wake up phrase **366'''** via the second stream **348** as part of the communication session between the first device **304** and the second device **306**.

[0071] In response to determining that command mode is the broadcast mode, the method **500** further includes monitoring the communication session for a spoken command, at **508**. In the example of FIG. **1**, the bridge device **102** monitors the first output stream **148** and the second output stream **152** for a spoken command. In the example of FIG. **3**, the first device **304** monitors the first stream **346** for a spoken command. In addition to monitoring the communication session, the first device **304** monitors input from the first microphone **310**.

[0072] In response to determining that command mode is the broadcast mode, the method **500** further includes detecting the spoken command, at **510**. In the example of FIG. **1**, the bridge device **102** detects the data corresponding to the command **164'''** in the first output stream **148**. In the example of FIG. **3**, the first device **304** detects a signal indicating the command **364''** from the first microphone **310**.

[0073] In response to determining that command mode is the broadcast mode, the method **500** further includes broadcasting the spoken command as part of the communication session, at **512**. In the example of FIG. **1**, the bridge device **102** broadcasts the data corresponding to the command **164''** to the second device **106** via the second input stream **150**. In the example of FIG. **3**, the first device **304** broadcasts data indicating the command **364'''** to the second device **306** via the second stream **348**.

[0074] In response to determining that command mode is the broadcast mode, the method **500** further includes performing one or more actions in response to the spoken command, at **514**. In the example of FIG. **1**, the bridge device **102** performs one or more actions, such as generating a meeting summary, retrieving content, retrieving information, issuing commands to a device, or a combination thereof based on the command **164**. In the example of FIG. **3**, the first device **304** performs one or more actions, as described above, in response to the command **364**.

[0075] In response to determining that command mode is the broadcast mode, the method **500** further includes broadcasting a reply to the spoken command as part of the communication session, at **516**. In the example of FIG. **1**, the bridge device **102** broadcasts the data corresponding to the reply **168'** to the first device **104** via the first input stream **146** and to the second device **106** via the second input stream **150**. In the example of FIG. **3**, the first device **304** broadcasts data indicating the reply **368'** to the second device **306** via the second stream **348**. The first device **304** also outputs an audio signal indicating the reply **368crv**) to the first speaker **308**.

[0076] In response to determining that command mode is the local mode, the method **500** further includes monitoring a local stream that is distinct from the communication session for a spoken command, at **518**. In the example of FIG. **2**, the bridge device **102** monitors the local output stream **208** for a spoken command. In the example of FIG. **4**, the first device **304** monitors input from the first microphone **310** for a spoken command.

[0077] In response to determining that command mode is the local mode, the method **500** further

includes detecting the spoken command, at 520. In the example of FIG. 2, the bridge device 102 detects the data corresponding to the command 212" in the local output stream 208. In the example of FIG. 4, the first device 304 detects a signal indicating the command 464" in input from the first microphone 310.

[0078] In response to determining that command mode is the local mode, the method 500 further includes performing one or more actions in response to the spoken command, at 522. In the example of FIG. 2, the bridge device 102 performs one or more actions, such as generating a meeting summary, retrieving content, retrieving information, issuing commands to a device, or a combination thereof based on the command 212. In the example of FIG. 4, the first device 304 performs one or more actions, as described above, in response to the command 464.

[0079] In response to determining that command mode is the local mode, the method 500 further includes outputting a reply to the command via a channel distinct from the communication session, at 524. In the example of FIG. 2, the bridge device 102 outputs the data corresponding to the reply 214' to the first device 104 via the local input stream 210. In the example of FIG. 4, the first device 304 outputs an audio signal indicating the reply 468' to the first speaker 308.

[0080] Thus, the method 500 enables processing of and reply to spoken commands. In particular the method 500 enables a device to determine how to respond to commands spoken during a communication session between devices. Therefore, the method 500 enables spoken commands to be used during a communication session between devices.

[0081] Referring now to FIG. 6, a block diagram illustrates a computing device 600 that is usable to implement the techniques described herein in accordance with one or more embodiments. For example, the computing device 600 illustrated in FIG. 6 could represent a client device or a physical server device. In some implementations, the computing device 600 corresponds the bridge device 102. As shown in FIG. 6, the computing device 600 can include one or more input/output devices, such as a network communication unit 608 that could include a wired communication component and/or a wireless communications component, which can be coupled to processing element 602. The network communication unit 608 can utilized any of a variety of standardized network protocols, such as Ethernet, TCP/IP, to name a few of many protocols, to effect communications between devices and comprise one or more transceiver(s) that utilize the Ethernet, power line communication (PLC), WiFi, and/or other communication methods.

[0082] The computing system 600 includes a processing element 602 that contains one or more hardware processors, where each hardware processor have a single or multiple processor cores. In one embodiment, the processing element 602 includes at least one shared cache that stores data (e.g., computing instructions) that are utilized by one or more other components of processing element 602. In a particular example, the shared cache corresponds to locally cached data stored in a memory for faster access by components of the processing elements 602. In one or more embodiments, the shared cache includes one or more mid-level caches, such as level 2 (L2), level 3 (L3), level 4 (L4), or other levels of cache, a last level cache (LLC), or combinations thereof. Examples of processors include, but are not limited to a central processing unit (CPU) a microprocessor. Although not illustrated in FIG. 6, in some embodiments, the processing element 602 includes one or more other types of hardware processing components, such as graphics processing units (GPU), application specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs), and/or digital signal processors (DSPs).

[0083] FIG. 6 illustrates that memory 604 is operatively coupled to processing element 602. In some embodiments, memory 604 corresponds to a non-transitory medium configured to store various types of data. In an illustrative example, the memory 604 includes one or more memory devices that comprise a non-volatile storage device and/or volatile memory. Volatile memory, such as random access memory (RAM), can be any suitable non-permanent storage device. The non-volatile storage devices can include one or more disk drives, optical drives, solid-state drives (SSDs), tap drives, flash memory, read only memory (ROM), and/or any other type memory

designed to maintain data for a duration time after a power loss or shut down operation. In certain instances, the non-volatile storage device is used to store overflow data if allocated RAM is not large enough to hold all working data. In some examples, the non-volatile storage device is also used to store programs that are loaded into the RAM when such programs are selected for execution. In the illustrated example, the memory **604** stores bridge instructions **612** and command processing instructions **614**. The bridge instructions **612** and the command processing instructions **614** are executable by the processor element **602** to perform one or more of the operations or methods described with respect to FIGS. 1-2 and FIG. 5. In particular, the bridge instructions **612** are executable by the processor element **602** to perform functions of a communication bridge. For example, in some implementations, the bridge instructions **612** are executable by the processor element **602** to process data received from the network communication unit **608** and to provide data to the network communication unit **608** in order to establish a communication session (e.g., a teleconference) between two devices. The command processing instructions **614** are executable by the processor element **602** to perform one or more actions described above in connection with responding to spoken commands. In some implementations, the command processing instructions **614** are executable to provide functionality of the natural language process backend module **120**, the meeting analytics module **122**, the natural language process frontend module **124**, or a combination thereof.

[0084] Persons of ordinary skill in the art are aware that software programs may be developed, encoded, and compiled in a variety computing languages for a variety software platforms and/or operating systems and subsequently loaded and executed by processing element **602**. In one embodiment, the compiling process of the software program transforms program code written in a programming language to another computer language such that the processing element **602** is able to execute the programming code. For example, the compiling process of the software program may generate an executable program that provides encoded instructions (e.g., machine code instructions) for processor element **602** to accomplish specific, non-generic, particular computing functions.

[0085] After the compiling process, the encoded instructions are then be loaded as computer executable instructions or process steps to processing element **602** from storage (e.g., memory **604**) and/or embedded within the processing element **602** (e.g., cache). Processing element **602** can execute the stored instructions or process steps in order to perform operations or process steps to transform the computing device into a non-generic, particular, specially programmed machine or apparatus. Stored data, e.g., data stored by a storage device, can be accessed by processing element **602** during the execution of computer executable instructions or process steps to instruct one or more components within the computing device **600**.

[0086] A user interface **610** can include a display, positional input device (such as a mouse, touchpad, touchscreen, or the like), keyboard, or other forms of user input and output devices. The user interface **610** can be coupled to processor element **602**. Other output devices that permit a user to program or otherwise use the computing device can be provided in addition to or as an alternative to network communication unit **608**. When the output device is or includes a display, the display can be implemented in various ways, including by a liquid crystal display (LCD) or a cathode-ray tube (CRT) or light emitting diode (LED) display, such as an OLED display. It should be noted that, in some embodiments, the computing device **600** comprises other components, such as sensors, powers sources, and/or analog-to-digital converters, not explicitly shown in FIG. 6.

[0087] Referring now to FIG. 7, a block diagram illustrates a computing device **700** that is usable to implement the techniques described herein in accordance with one or more embodiments. For example, the computing device **700** illustrated in FIG. 7 could represent a client device or a physical server device. In some implementations, the computing device **700** corresponds the first device **104**, the second device **106**, the first device **304**, the second device **306**, or a combination thereof. As shown in FIG. 7, the computing device **700** includes a processor element **702** coupled

to a user interface **710**, a memory **704**, and a network communication unit **708**. In some implementations, the processor element **702**, the user interface **710**, the memory **704**, and the network communication unit **708** include one or more devices described above with reference to the processor element **602**, the user interface **610**, the memory **604**, and the network communication unit **608**. The processor element is further coupled to a speaker **722** and a microphone **724**.

[0088] In the illustrated example, the memory **704** stores endpoint instructions **712** and command processing instructions **714**. In some implementations (e.g., some implementations in which endpoints are configured to communicate via a bridge device), the memory **704** does not store the command processing instructions **714**. The bridge endpoint instructions **712** and the command processing instructions **714** are executable by the processor element **602** to perform one or more of the operations or methods described with respect to FIGS. 1-5. In particular, the endpoint instructions **712** are executable by the processor element **702** to perform functions of a communication endpoint (e.g., an audio input/output device). For example, in some implementations, the endpoint instructions **712** are executable by the processor element **702** to establish a communication session with another endpoint (e.g., a bridge facilitated communication session or a peer to peer communication session). During the communication session, the endpoint instructions **712** are executable by the processor element **702** to process audio signals received from the microphone **724** into audio data and to provide the audio data to the network communication unit **708** to be communicated to another device. The endpoint instructions **712** are further executable by the processor element **702** to receive additional audio data from the network communication unit **708** and to output an additional audio signal to the speaker **722** based on the additional audio data. The communication session may be facilitated by a bridge device or may be a peer to peer communication session.

[0089] The command processing instructions **714** are executable by the processor element **702** to perform one or more actions described above in connection with transmitting and responding to spoken commands. The microphone **724** is configured to detect the spoken commands and various associated phrases. In some implementations, the command processing instructions **714** are executable to provide functionality of the first wake up detector module **130**, the first echo canceller module **132**, the first mixer module **134**, the first controller module **136**, the second echo canceller module **138**, the second wake up detector module **140**, the second controller module **142**, the second mixer module **144**, the first wake up detector module **330**, the first echo canceller module **332**, the first mixer module **334**, the first controller module **336**, the second controller module **342**, the second mixer module **344**, the second wake up detector module **340**, the second echo canceller module **338**, the first natural language process backend module **320**, the first meeting analytics module **322**, the first natural language process frontend module **324**, or a combination thereof.

[0090] As discussed above, the systems and methods described above with reference to FIGS. 1-7 enable a system to process and respond to spoken commands during a communication session (e.g., a teleconference). Accordingly, a user can use spoken commands in lieu of or in addition to remote control commands to interact with the system during the communication session. Spoken commands may be more convenient as compared to remote control entered commands. Additionally, a manufacturer may create a system that supports many more commands than an associated remote control device supports.

[0091] At least one embodiment is disclosed and variations, combinations, and/or modifications of the embodiment(s) and/or features of the embodiment(s) made by a person having ordinary skill in the art are within the scope of the disclosure. Alternative embodiments that result from combining, integrating, and/or omitting features of the embodiment(s) are also within the scope of the disclosure. Where numerical ranges or limitations are expressly stated, such express ranges or limitations are understood to include iterative ranges or limitations of like magnitude falling within

the expressly stated ranges or limitations (e.g., from about 1 to about 10 includes, 2, 3, 4, etc.; greater than 0.10 includes 0.11, 0.12, 0.13, etc.). The use of the term “about” means $\pm 10\%$ of the subsequent number, unless otherwise stated.

[0092] Use of the term “optionally” with respect to any element of a claim means that the element is required, or alternatively, the element is not required, both alternatives being within the scope of the claim. Use of broader terms such as comprises, includes, and having is understood to provide support for narrower terms such as consisting of, consisting essentially of, and comprised substantially of. Accordingly, the scope of protection is not limited by the description set out above but is defined by the claims that follow, that scope including all equivalents of the subject matter of the claims. Each and every claim is incorporated as further disclosure into the specification and the claims are embodiment(s) of the present disclosure.

[0093] It is to be understood that the above description is intended to be illustrative, and not restrictive. For example, the above-described embodiments are useable in combination with each other. Many other embodiments will be apparent to those of skill in the art upon reviewing the above description. The scope of the invention therefore should be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. It should be noted that the discussion of any reference is not an admission that it is prior art to the present invention, especially any reference that has a publication date after the priority date of this application.

Claims

1. A method of responding to spoken commands in a communication session, the method comprising: receiving a first signal from a first device during a communication session between the first device and a second device, the first signal indicating a wake up message; receiving a second signal from the first device, after the first signal, indicating a spoken command; when in a first command mode, broadcasting the wake up message and the spoken command to the second device; when in a second command mode, not broadcasting the wake up message and the spoken command to the second device; and performing an action based on the spoken command while maintaining the communication session between the first device and the second device.
2. The method of claim 1, wherein the wake up message includes a wake up element and a command element; and further comprising: selecting one of the first command mode or the second command mode based on the command element.
3. The method of claim 1, further comprising processing the spoken command using a natural language process module to detect data corresponding to the spoken command, wherein the data detected by the natural language process module includes communication session context, content of the spoken command, information identifying a meeting object, or any combination thereof.
4. The method of claim 3, wherein the meeting object includes a participant in the communication session, a list of user profiles, a first list of locations, a second list of devices, a third list of supported spoken commands, or any combination thereof.
5. The method of claim 1, wherein the spoken command includes a request for information, a media content item, to control a part of the communication session, to control the first device in the communication session, to control the second device in the communication session, or any combination thereof.
6. The method of claim 5, wherein the request to control the second device includes a request to activate the second device, a request to adjust a setting of the second device, or both.
7. The method of claim 1, wherein the action includes at least one of: (i) creating a meeting summary associated with the communication session; (ii) requesting weather information for a location from a weather service; (iii) controlling a lighting device in the communication session; and (iv) determining if a user who created the spoken command is authorized to initiate the spoken

command; and (v) displaying media content on a graphical user interface of the first device based on a keyword in the spoken command.

8. The method of claim 1, further comprising selecting one of the first command mode or the second command mode for the first device based on the wake up message, independent of a selected command mode of the second device.

9. An apparatus comprising: a processor; and a memory storing instructions that, when executed by the processor, cause the processor to: receive a first signal from a first device during a communication session between the first device and a second device, the first signal indicating a wake up message; receive a second signal from the first device, after the first signal, indicating a spoken command; perform an action based on the spoken command while maintaining the communication session between the first device and the second device; generate a reply corresponding to the action; when in a first command mode, broadcast the reply to both the first device and the second device; and when in a second command mode, broadcast the reply to only the first device.

10. The apparatus of claim 9, wherein the processor is to process the spoken command using a natural language process module to detect data corresponding to the spoken command, wherein the apparatus further includes a first controller module that, when in the first command mode, inserts the data detected by the natural language process module into a first output stream between the first device and the second device.

11. The apparatus of claim 10, wherein the data detected by the natural language process module includes communication session context, content of the spoken command, information identifying a meeting object, or any combination thereof.

12. The apparatus of claim 9, wherein the spoken command includes a request for information, a media content item, to control a part of the communication session, to control the second device in the communication session, or any combination thereof.

13. The apparatus of claim 12, wherein the request to control the part of the communication session includes a request to connect to a third device.

14. The apparatus of claim 12, wherein the request to control the second device includes a request to activate the second device, a request to adjust a setting of the second device, or both.

15. The apparatus of claim 9, wherein the action includes at least one of to: (i) generate a meeting summary associated with the communication session; (ii) request weather information for a location from a weather service; (iii) control a lighting device in the communication session; and (iv) determine if a user who created the spoken command is authorized to initiate the spoken command; and (v) display media content on a graphical user interface of the first device based on a keyword in the spoken command.

16. A non-transitory computer-readable medium containing instructions that when executed cause a processor to: receive a first signal from a first device during a communication session between the first device and a second device, the first signal indicating a wake up message; set a chosen command mode of a plurality of command modes; receive a second signal indicating a spoken command; and perform an action based on the spoken command and the chosen command mode while maintaining the communication session between the first device and the second device.

17. The non-transitory computer-readable medium of claim 16, wherein to select the chosen command mode of the plurality of command modes includes to select the chosen command mode based on a lack of command element in the wake up message.

18. The non-transitory computer-readable medium of claim 16, wherein the wake up message includes a wake up element and a command element, and wherein to select the chosen command mode of the plurality of command modes includes to select the chosen command mode based on the command element.

19. The non-transitory computer-readable medium of claim 16, wherein the chosen command mode indicates whether the second device receives a broadcast of the wake up message.

20. The non-transitory computer-readable medium of claim 16, wherein the chosen command mode indicates whether the second device receives a reply corresponding to the performed action.
